

Physics-Informed State Reconstruction and Event-Source Inference From Sparse PMUs Under Limited Observability

JORGE LUIS MAYORGA TABORDA¹, YESSICA AFRICANO RODRIGUEZ¹, MASOUD BARATI²,
PAYMAN DEGHANIAN³, AND DAVID FELIPE CELEITA RODRÍGUEZ⁴

¹Universidad de los Andes, Bogotá, Colombia

²Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA

³Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA

⁴Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

ABSTRACT Sparse phasor measurement units leave two different questions unresolved: whether the unmeasured voltage field can be reconstructed and whether the physical sources of an event can be distinguished. We separate those questions with a hybrid differential-algebraic model. Functional observability governs the requested voltage target; finite-horizon separation between candidate response manifolds governs event cardinality and source support. We test this distinction on the IEEE 39-bus system with eight voltage/current PMUs. A causal model-based estimator reconstructs the 31 unmeasured bus voltages with 0.009634% mean relative error under matched operation. A separate physical likelihood evaluates 137 load-intervention hypotheses, comprising no event, every single source, and every two-source support. On 3488 fresh nonlinear trajectories, exact-support Top-1 ranges from 24.4% for weak pairs to 100.0% for finite pairs. The weak cases reveal the central limitation: posterior evidence can identify that two sources are active before it can identify which two. Profiled response distance explains this ambiguity better than support-only geometry, while longer observation increases separation without removing severe signature collinearity. The results support matched-model state reconstruction and simultaneous load-source inference. They do not establish a multi-family joint estimator, changing-PMU operation, unknown-onset inference, or field transfer.

INDEX TERMS dynamic state estimation, event detection, functional observability, limited observability, phasor measurement units, physics-informed inference, source localization.

I. INTRODUCTION

PHASOR measurement units (PMUs) reveal electromechanical transients through synchronized voltage and current phasors, yet transmission networks instrument only a fraction of their buses and branches [1], [2]. An event may start outside the measured region and still alter every available PMU. Missing frames and corrupted channels create a second problem because they can resemble a physical transient. A useful diagnosis must therefore answer separate questions: what is the hidden network state, what changed physically, and which measurements can still be trusted?

Dynamic estimators combine network models with synchrophasors [2], [3]; sparse-PMU classifiers identify events and faulted assets [4], [5]; and recent learning methods reconstruct hidden trajectories before localization [6], [7]. These capabilities do not imply one another. In particular, a low state-reconstruction error says nothing about whether two

candidate sources produce distinguishable measurements.

This distinction changes how the problem must be evaluated. A generation change may be visible throughout the network while its source remains ambiguous. A dropped PMU identifies the failed measurement device, yet the lost channel can make a simultaneous physical event harder to localize. Treating both cases as entries in one multiclass label hides the relevant failure mechanism and mixes incompatible denominators.

Our central idea is to separate target recovery from source separation inside the same physical model. A candidate event specifies a family, a sparse support, an onset, and signed severity. Measurement availability and corruption remain separate variables. Each candidate generates a counterfactual DAE response. After admissible state and integrity directions are profiled, the distance between candidate response manifolds measures the information available for source dis-

crimination [8], [9]. Zero distance means that two candidates intersect; a small positive distance means that they are distinct but difficult to separate with the available horizon and noise.

We test that argument in two stages. The state study asks what eight voltage/current PMUs recover, how the estimate fails when its operating point is stale, and whether a constrained physical recentering repairs that failure. The event study then evaluates all zero-, one-, and two-source load supports without fitting a bus-label classifier. A retrospective geometry audit and a longer-horizon diagnostic explain the weak-source failures. Finally, an earlier supervised ANDES model provides a contrasting test: it performs well when a source appears in training but collapses under complete source holdout. These studies use separate targets and denominators.

The contributions are:

- 1) a hybrid DAE formulation that keeps physical interventions and measurement integrity compositional, and represents simultaneous sources by a sparse support rather than a new class label;
- 2) a finite-horizon analysis that separates functional state recovery, sparse event injectivity, and the conditioning supplied by additional PMUs or observation time; and
- 3) a prospective 137-hypothesis load-source test that connects exact-support recovery to response-manifold geometry, together with independent state-reconstruction and source-holdout evidence.

The evidence yields one consistent picture. Sparse PMUs can recover the requested voltage field under a matched model, but operating-point error must be corrected physically. For simultaneous load events, source severity controls whether the posterior resolves only cardinality or the exact support. Longer observation adds information, yet nearly collinear signatures remain difficult. The evaluated method therefore supports matched load interventions with at most two sources. It does not yet solve the full multi-family problem shown in Fig. 1.

II. PRIOR WORK AND RESEARCH GAP

A. STATE, ALGEBRAIC VARIABLE, AND UNKNOWN-INPUT ESTIMATION

Power-system dynamic state estimation combines machine, control, load, and network models with synchronized measurements; model uncertainty and measurement availability remain persistent constraints [2]. Robust filters address non-Gaussian measurement errors [10], [11], whereas joint input/state estimators recover exogenous inputs in linear systems [12], nonlinear DAEs [3], [13], and event-triggered or distributed power networks [14], [15]. These estimators do not define a multi-family physical-source bank with a simultaneous integrity state.

Learning methods reconstruct states from sparse measurements by encoding graph structure or power-flow relations. Examples combine graph learning with physics-based estimates [16], outlier-resistant estimation and outage identification [17], or topology-aware reconstruction under PMU

failures and bad data [18]. Robust reconstruction does not, by itself, identify the physical source.

B. SPARSE-PMU EVENT AND SOURCE INFERENCE

PMU event identification uses subspaces, engineered features, transfer learning, and graph models [19]–[22]. Label-light field-stream detection and scarce-label transfer target event presence or family [23], [24]; physical-source recovery is distinct. Prior localization spans joint classification/localization [4], sparse-PMU fault, generation-loss, and outage methods [5], [25]–[27], group-sparse physical dictionaries [28], and topology-constrained graph losses [29].

Model-based outage inference provides the closest classical comparison. Phasor angles support line-outage detection [30]; uncertain pre-event state can be marginalized jointly with outage identity and sensor placement [31]; and sequential likelihoods handle stochastic operating variation [32], [33]. Matched-subspace detectors profile linear nuisance through a generalized likelihood ratio [9]. These are the likelihood and nuisance-elimination precedents used here. Our load contract instead uses a sparse voltage/current response dictionary and known onset; it is not a quickest-detection comparison.

Open-set recognition differs from source holdout. Siamese boundaries and generative evidence reject events outside learned classes [34]–[36]. Source generalization instead ranks a physically enumerable but unlabeled asset as a counterfactual source.

The pmuBAGE collection supports standardized analytics but does not certify source holdout [37]. Integrity is a separate variable: robust recovery distinguishes excursions from outliers [38], while utility examples include missing, locked, quantized, and timing-affected synchrophasors [39].

The unresolved issue is the evaluation contract. New replicas from trained assets measure interpolation, not source generalization; nuisance variation can also alter conclusions [40]. Reconstruction followed by localization is prior art [6], [7], but not under the combined held-source, diagnosability, integrity, and mismatch tests used here.

C. OBSERVABILITY, DIAGNOSABILITY, AND SENSOR VALUE

Functional observability asks whether a requested function of state can be inferred without full-state observability [41], [42]; nonlinear extensions require the observation fibers to determine the requested functional locally, not merely a favorable Jacobian at one point [43]. Unknown-input observability instead asks whether exogenous inputs can be reconstructed from output histories [44]; fault detectability asks whether a fault can be separated while disturbances are decoupled [45]. Sparse strong observability extends the question to corrupted sensor sets [46], [47], and attack identification characterizes when adversarial input supports are distinguishable from system behavior [48]. Eriksson *et al.* quantify pairwise diagnosability for stochastic linear descriptor models [8]. Likelihood inference with singular first-order information also has es-

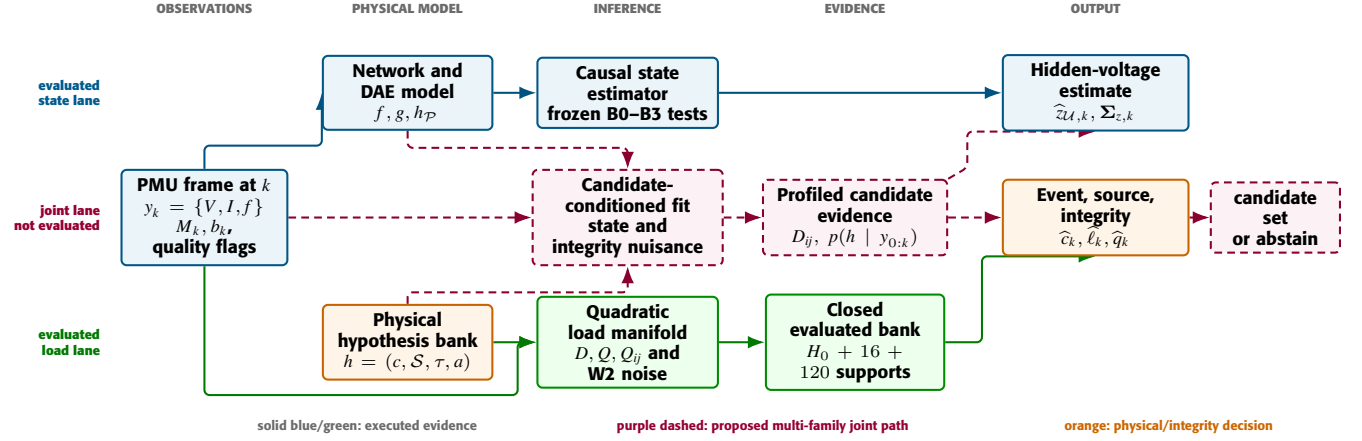


FIGURE 1. Scientific argument and validation boundary. The evaluated state path asks what the sparse PMUs recover. The evaluated load-event path asks which candidate supports the measurements distinguish. The dashed middle path is the intended multi-family integration; it remains outside the present evidence.

established nonregular asymptotics [49]; therefore, a vanishing projected tangent is not by itself a universal impossibility result. PMU placement based on empirical observability Gramians treats sensor value as task dependent [50], [51]. For event sources, Jamei *et al.* derive placement-dependent fault-localization limits [52]; Conte *et al.* partition a radial grid into indistinguishable line clusters [53]; and Yildiz and Abur optimize sparse placement for event detection and identification [54]. These results distinguish hidden-state reconstruction, input reconstruction, and pairwise source separation. They also rule out a general optimal-placement claim from one prescribed eight-PMU set.

D. REMAINING TECHNICAL GAP

Table 1 shows why one accuracy value cannot close the gap. The state path must specify the requested function, and the event path must expose indistinguishable sources before any classifier is trusted. A source-generalization test must also remove the physical asset from every source-labeled fitting step. Under that contract, learning is useful only if it corrects model discrepancy or reduces candidate-search cost without memorizing asset identity.

The experiments cover only part of this intersection. PowerDynamics supports hidden-voltage estimation, physical re-centering, and prospective load-source inference. ANDES documents what a supervised representation loses under source holdout. A common multi-family experiment with state and integrity nuisance remains future work.

III. STATE, EVENT, AND SOURCE PROBLEM

A. ONE PHYSICAL MODEL, TWO INFERENCE QUESTIONS

Let $x(t)$ and $z(t)$ denote differential and algebraic states, $q(t)$ the network mode, and θ the operating parameters. A physical candidate is

$$h = (c, \mathcal{S}, \tau, a, d), \quad (1)$$

where c is the event family, $\mathcal{S}$ its bus or branch support, τ the onset, a the signed severity, and d the temporal profile. The

hybrid plant and measurement boundary are

$$\dot{x} = f_h(x, z, u; \theta) + w_x, \quad 0 = g_h(x, z, u; \theta), \quad (2)$$

$$y_k = M_k[h_{\mathcal{P}_k}(x_k, z_k, q_k; \theta) + b_k] + v_k. \quad (3)$$

The event enters the physical equations through h . In contrast, M_k records channel availability and b_k records measurement corruption. A physical event and a failed PMU can therefore coexist without becoming a new event class.

The network map uses $I = Y_{\text{bus}}V$ and $S_i = V_i I_i^*$. A fault changes a shunt, a line outage changes Y_{bus} , and generation or load events change the corresponding injections. PMU currents use the instrumented branch terminal and orientation. Supplementary Appendix B gives the AC power equations, transformer and terminal-current conventions, index-one reduction, and frequency/ROCOF definitions. Those details determine the likelihood, but they do not change the two inference questions posed here.

The first question asks for a function of state. With PMUs at

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}, \quad (4)$$

the requested target is the complex voltage at the remaining 31 buses. The state estimator uses rectangular voltage and controlled terminal-current channels. The second question asks which physical candidate generated the observed response. For the evaluated load family,

$$\mathcal{S} \subseteq \mathcal{L}, \quad |\mathcal{L}| = 16, \quad K = |\mathcal{S}| \leq 2, \quad (5)$$

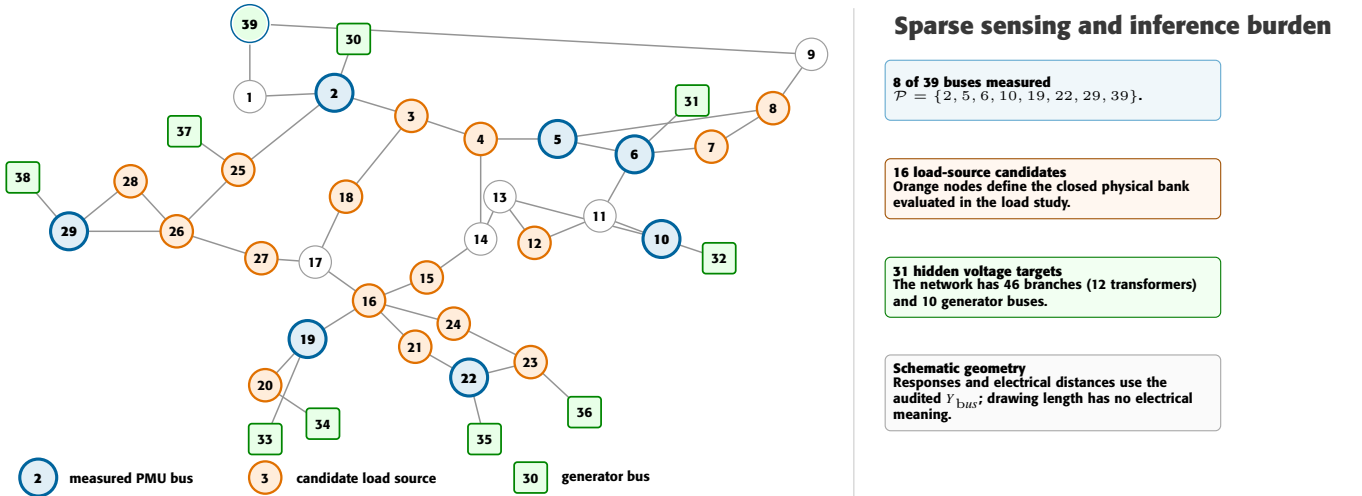
which yields one no-event hypothesis, 16 single-source hypotheses, and 120 two-source supports.

B. PHYSICAL EVENTS AND MEASUREMENT INTEGRITY

The historical event bank has eight abnormal labels, but the label number is not the scientific variable. Table 2 rewrites the bank as four physical interventions, two integrity states, and

TABLE 1. Research gap by methodological family. The full paper-by-paper capability matrix is reported in Supplementary Appendix A.

Family	Established capability	Remaining issue for this problem	Role here
Dynamic state and unknown-input estimation [3], [12], [13]	Reconstructs states or unknown inputs from model and PMU histories	Does not usually define a closed physical event-source bank with compositional integrity	Defines the state-reconstruction path and its nuisance variables
Learned event localization [4], [5], [29]	Detects and localizes represented event families under sparse sensing	Source-label training can confound asset memorization with source generalization	Supplies the supervised baseline and held-source stress test
Model-based event inference [30]–[32]	Ranks physical outage hypotheses and can marginalize operating uncertainty	Published contracts are event-specific and do not establish the present simultaneous-load problem	Motivates physical candidate likelihoods and nuisance profiling
Observability and placement [43], [52], [54]	Characterizes state recovery or event separation for a declared sensing task	Rank alone does not quantify finite-horizon support ambiguity	Supplies the target-recovery and diagnosability tests
This work	Reconstructs a requested hidden-voltage field and ranks all load supports with $K \leq 2$	The multi-family state/event/integrity posterior remains untested	Connects separate evaluated paths through one information-based formulation

**FIGURE 2.** Sensing and candidate geometry on the IEEE 39-bus system. Blue circles mark the eight PMU buses, orange circles the 16 evaluated load-source buses, and green boxes generator buses. The drawing preserves the audited 46-branch connectivity but does not encode geographic or electrical distance.

two prescribed compositions. This representation matters because a missing packet identifies an integrity source directly while reducing the information available for physical-source inference.

The desired decision consists of the hidden-voltage estimate, its uncertainty, the physical family and support, and the integrity state. These components use different correctness tests. Hidden-voltage relative error is averaged over a complete trajectory and the 31 requested buses. Event detection, family classification, exact-support recovery, and integrity localization retain their own populations. A joint decision is correct only when every applicable component is correct.

Causality and generalization are also explicit. A causal estimate at time k may use only $y_{0:k}$; fixed-lag or complete-window results are retrospective. A held-source test removes the physical asset from every source-labeled fitting and tun-

ing step. Confidence intervals use trajectories, event parents, or rebuilt plants as independent units, never individual 30-Hz samples. Supplementary Appendix C records the exact metrics and event operators.

IV. WHAT SPARSE MEASUREMENTS CAN DETERMINE

The analysis separates two questions before an estimator is chosen. The first asks whether the measured history determines the requested hidden-voltage function. The second asks whether it distinguishes competing event supports. Both depend on the sensing geometry, but they fail for different reasons.

TABLE 2. Event ontology and historical ANDES inventory. Each source configuration has five replicas. Physical and integrity sources remain separate outputs.

ID	Operational meaning	Physical intervention	Physical source	Integrity state/source	Records
0	Normal operation	none	none	nominal	5
1	Bus fault	fault admittance	bus	nominal	195
2	Line outage	topology removal	branch	nominal	170
3	Generation change	active/reactive injection	generator bus	nominal	50
4	Load change	active/reactive withdrawal	load bus	nominal	90
5	Missing data	none	none	dropout / PMU	40
6	Generation change with missing data	generation change	generator bus	dropout / PMU	50
7	Bad data	none	none	corruption / PMU channel	40
8	Prescribed mixed disturbance	generation plus load changes	declared generator bus	measurement disturbance / PMU	50
Total					690

A. RECOVERING A REQUESTED VOLTAGE FUNCTION

Linearizing the index-reduced DAE around a verified equilibrium gives

$$\delta x_{k+1} = A_d \delta x_k + w_k, \quad \delta y_k = C \delta x_k + v_k, \quad (6)$$

$$\delta z_{\mathcal{U},k} = L_{\mathcal{U}} \delta x_k.$$

Here $\delta x_k \in \mathbb{R}^{114}$, the 32 measured channels form δy_k , and the 62 rectangular voltage components at the 31 unmeasured buses form the target. For a horizon of N transitions, let

$$\mathcal{O}_N = \text{col}(C, CA_d, \dots, CA_d^N), \quad L_r = L_{\mathcal{U}} A_d^r. \quad (7)$$

Proposition 1 (Finite-horizon functional reconstruction): The noise-free measurement history through N uniquely determines the requested voltage at time $r \leq N$ if and only if

$$\ker \mathcal{O}_N \subseteq \ker L_r. \quad (8)$$

The condition is weaker than full-state observability: unobservable state directions are harmless when they do not change the requested voltage. Under Gaussian noise, the same row-space condition yields the minimum-variance linear unbiased target estimate; its covariance depends on singular values and target alignment, not rank alone [41], [42]. Supplementary Appendix D gives the proof, the covariance expression, and the nonlinear constant-rank qualification [43].

The numerical audit illustrates why that qualification matters. The original implementation formed $\mathcal{O}_N^T \mathcal{O}_N$ and applied a relative eigenvalue threshold. Its pseudo-rank rises and then falls as samples are added, which an exact observability rank cannot do. Figure 3 therefore separates two conclusions. Longer history improves numerical conditioning, but the tested computation does not certify full-state or exact functional observability. The nonlinear estimator is evaluated independently.

The bus-level residual from this audit was frozen before the nonlinear TEST trajectories were scored. Its association with empirical bus error is used only as an exploratory theory-to-data check because the projector remains threshold sensitive. An SVD-based, noise-tied rank audit is still required for a structural certificate.

B. DISTINGUISHING EVENT SUPPORTS

Source inference introduces a different nuisance space. Under candidate h , stack the available measurements over H samples as

$$Y_H = s_{h,H} + \mathcal{O}_{h,H} \xi_h + \mathcal{E}_{S,H} \beta_h + \varepsilon_H, \quad (9)$$

where $s_{h,H}$ is the physical response, ξ_h an admissible initial-state deviation, and $\mathcal{E}_{S,H} \beta_h$ corruption on integrity support S . Each candidate therefore defines an affine response manifold. For candidates i and j , their profiled separation is

$$D_{ij,S}(H) = \frac{1}{2} \min_{\xi_i, \xi_j, \beta} \|s_{i,H} + \mathcal{O}_{i,H} \xi_i - s_{j,H} - \mathcal{O}_{j,H} \xi_j + \mathcal{E}_{S,H} \beta\|_{R_H^{-1}}^2. \quad (10)$$

This distance profiles the nuisance directions that either candidate can use to explain the same measurements. It specializes descriptor-model diagnosability and matched-subspace testing to the PMU event bank [8], [9].

Proposition 2 (Local candidate indistinguishability): If $D_{ij,S}(H) = 0$, the two candidate manifolds intersect. At an intersection they induce the same Gaussian measurement law, so no rule can distinguish them uniformly over the declared nuisance set.

A positive distance establishes local separation, not an easy decision. Small distance or near-collinear signatures amplify noise and delay support resolution. Supplementary Appendix D proves the proposition, treats unknown sparse corruption, and shows that adding a conditionally independent PMU cannot reduce the profiled margin when the nuisance contract is fixed.

For persistent load events, subtract the common no-event trajectory and write the T -sample response as

$$Y_T = \mathcal{H}_T a + \varepsilon_T. \quad (11)$$

Let $\mathcal{A}_{2K}$ contain vectors with at most $2K$ nonzero entries.

Proposition 3 (Finite-horizon sparse event injectivity): Every event with at most K active sources is uniquely determined in the noise-free model if and only if

$$\ker \mathcal{H}_T \cap \mathcal{A}_{2K} = \{\mathbf{0}\}. \quad (12)$$

Stable amplitude estimation also requires a well-conditioned support Gramian.

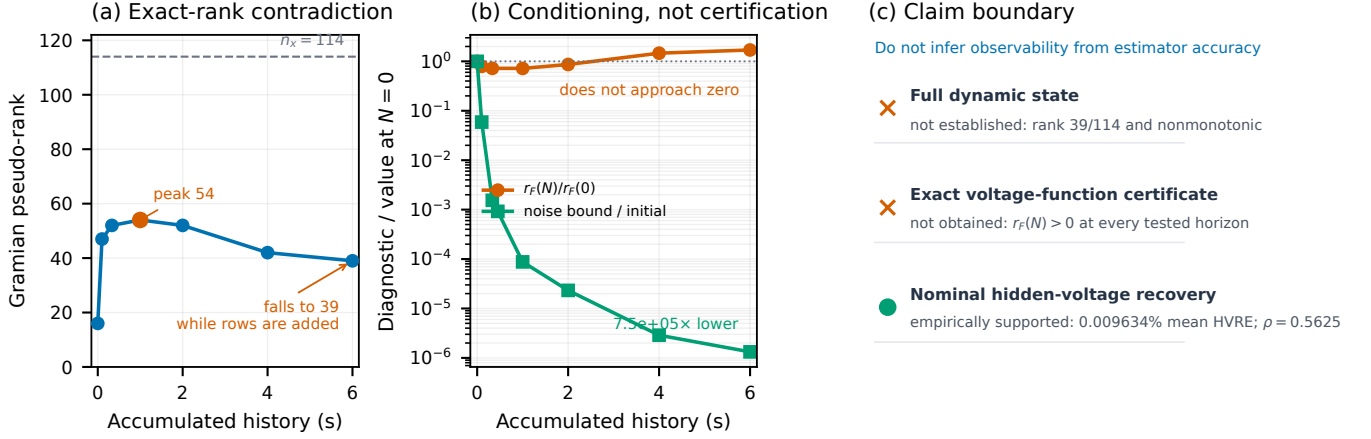


FIGURE 3. Structural audit and claim boundary. Adding history lowers the target-noise bound, but the relative-threshold pseudo-rank is nonmonotone and the target residual remains nonzero. The audit therefore motivates a conditioning analysis; it does not prove full-state or exact hidden-voltage observability.

The proposition distinguishes uniqueness from finite-information difficulty. With prefix-consistent causal whitening, adding samples cannot reduce the profiled separation between two fixed supports. Separation grows when their steady signatures differ and can saturate when only a finite transient distinguishes them. The experiments therefore ask two questions in order: does the posterior identify how many sources are active, and does the remaining separation identify which sources? Full proofs and the temporal cases appear in Supplementary Appendix D.

V. TWO EVALUATED INFERENCE PATHS

Figure 1 separates the implemented paths from the intended joint model. The blue path reconstructs a requested voltage function under one physical mode. The green path compares load-event supports through simulated physical responses. They share the DAE and PMU operator, but they do not share a fitted posterior in the present experiments.

A. CAUSAL STATE RECONSTRUCTION

The state experiment changes one information source at a time. B0 returns the equilibrium voltage. B1 conditions that equilibrium on the current PMU frame. B2 propagates the state causally and then applies the PMU correction. B3 is a retrospective fixed-lag smoother used only to test whether later measurements add useful information. This sequence isolates where each improvement comes from.

For the local model in (6), B2 uses the standard Kalman recursion [55]:

$$\begin{aligned}\hat{x}_{k|k-1} &= A_d \hat{x}_{k-1|k-1}, \\ P_{k|k-1} &= A_d P_{k-1|k-1} A_d^T + Q, \\ K_k &= P_{k|k-1} C^T (C P_{k|k-1} C^T + R)^{-1}, \\ \hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k \nu_k.\end{aligned}\quad (13)$$

The hidden-voltage estimate is $z_{U,0} + L_U \hat{x}_{k|k}$. Only measurements through k enter B2. B3 follows the Gaussian backward recursion [56] and is explicitly retrospective.

Two checks separate point accuracy from uncertainty. First, a temperature map fitted on independent calibration trajectories rescales the two rectangular voltage components without changing the state estimate. Second, an observed-innovation statistic tests whether the nominal model has become inadequate. Neither operation repairs the physical model. Their complete equations and negative discrepancy ablations appear in Supplementary Appendix E.

When the operating point is stale, the physical-recentering estimator rebuilds a feasible equilibrium instead of shifting measured voltages by an unconstrained offset. It solves

$$(\hat{V}, \hat{d}) = \arg \min_{V, d} \|e_y(V)\|_2^2 + \gamma^2 \|g_{AC}(V, d)\|_2^2 + \lambda q_d^{-2} \|d\|_2^2, \quad (14)$$

where e_y uses only measured voltage and terminal-current phasors, g_{AC} enforces AC balance and bus conventions, and d collects admissible load and generation changes. Development data fix the weights and continuation rule before TEST. Hidden voltages and the true nuisance vector never enter the solve.

B. PHYSICAL LIKELIHOOD FOR LOAD-EVENT SUPPORTS

The event path begins with a physical question: what would the measured PMUs have recorded if a candidate load bus had changed? For each of the 16 buses, centered nonlinear simulations form the unit response

$$D_g = \frac{y_g(+\epsilon) - y_g(-\epsilon)}{2\epsilon}, \quad \epsilon = 0.005. \quad (15)$$

Symmetric simulations also estimate source self-curvature Q_i and pairwise interaction Q_{ij} . The mean response for support S is

$$\mu_S(a) = \sum_{i \in S} (a_i D_i + a_i^2 Q_i) + \sum_{\substack{i, j \in S \\ i < j}} a_i a_j Q_{ij}. \quad (16)$$

The mixed term matters physically: two simultaneous interventions need not equal the sum of two isolated responses.

After subtracting the normal mean, the residual follows $r = \mu_S(a) + \varepsilon$ under the frozen Gaussian covariance. Each support receives normalized evidence by integrating over its active amplitudes,

$$Z_S = \int \mathcal{N}(r; \mu_S(a), \Sigma) \prod_{i \in S} p(a_i) da_S, \\ p(S | r) \propto Z_S \frac{p(|S|)}{\binom{16}{|S|}}. \quad (17)$$

Gauss–Hermite quadrature evaluates every support; independent Gauss–Kronrod calculations check representative two-dimensional integrals. Priors, covariance estimation, quadrature order, and calibration checks are documented in Supplementary Appendix E.

The posterior is read in a fixed order. First, $p(K | r)$ asks whether no source, one source, or two sources are active. Conditional support ranks then ask which buses are involved. Amplitude intervals answer how large the changes are. This order is essential under limited observability because cardinality can be well supported while several source pairs remain plausible.

The prospective experiment evaluates all 137 supports at a known onset and fixed PMU set. It conditions on the nominal initial state and assumes no corrupted channels. Thus it evaluates the sparse load specialization of Proposition 3, not the full state-and-integrity nuisance model in (9).

C. BOUNDARY OF THE JOINT FORMULATION

The complete candidate-conditioned model would fit state and integrity nuisance for each physical hypothesis and normalize the resulting evidence. A learned component could then correct candidate residuals or reduce the search, but it must not consume source identity as a shortcut. The required common-contract ablation is learned-only, physics-only, physics plus learned discrepancy, and diagnosability-aware inference with abstention. None of those integrations is presented as an executed result here; Supplementary Appendix E records the full generative objective so the proposed extension is unambiguous.

VI. EXPERIMENTAL QUESTIONS AND FROZEN EVIDENCE

The experiments follow the logic of the paper rather than the order in which the software was developed. Each research question has its own independent unit, frozen selection boundary, and claim limit (Table 3). The complete manifests, seeds, acceptance gates, simulator versions, and secondary campaigns appear in Supplementary Appendix F.

A. RQ1 AND RQ2 STATE-RECONSTRUCTION CONTRACTS

The primary plant is the package-defined IEEE 39-bus PowerDynamics benchmark [57]. Before scoring estimator error, we verified the power-flow translation, terminal-current orientation, equilibrium, local modes, nonlinear trajectories, and first-order response. These checks establish numerical

consistency with the implemented benchmark; they do not make the tutorial model an authoritative external grid.

RQ1 compares B0–B3 on the same 3-s trajectories. Model parameters and covariance choices are fixed before TEST, and the 31 requested buses exclude all eight observed buses. Separate calibration trajectories fit uncertainty corrections. RQ2 then rebuilds the plant under seven physical mismatch families while keeping the nominal estimator frozen. A distinct held-out operating-point campaign tests the constrained recentering in (14). Hidden voltages are revealed only after each solve, and rejected solves remain in the planned denominator.

B. RQ3 SIMULTANEOUS LOAD-SOURCE CONTRACT

Development data select the normal covariance and likelihood before the prospective support test. The final bank freezes D_i , Q_i , Q_{ij} , the priors, onset, and quadrature, then generates new nonlinear trajectories for H_0 , every single source, and every source pair. Event identifiers do not overlap the historical development manifests. Repeated noise realizations produce separate decisions but are not counted as new physical-system trajectories.

The primary outcome is exact support, evaluated only after cardinality. A retrospective weak-regime audit asks which geometric distance explains errors without refitting the likelihood. A separate 120-frame study checks physical-manifold fidelity, information accumulation, and runtime. Because that study uses only one operating point and 32 validation paths, it is a diagnostic rather than a prospective recovery result.

C. RQ4 HISTORICAL BASELINE AND EVENT SEMANTICS

The ANDES campaign [58] preserves the earlier physics-guided hierarchy as a baseline. Its compact representation uses local voltage, angle, frequency, and availability information; its expanded representation adds multiscale dynamics, adaptive residuals, and simultaneous PMU contrasts. A deterministic availability gate handles observed dropout. The ensemble is not DAE constrained, and its leaf votes are not treated as calibrated probabilities.

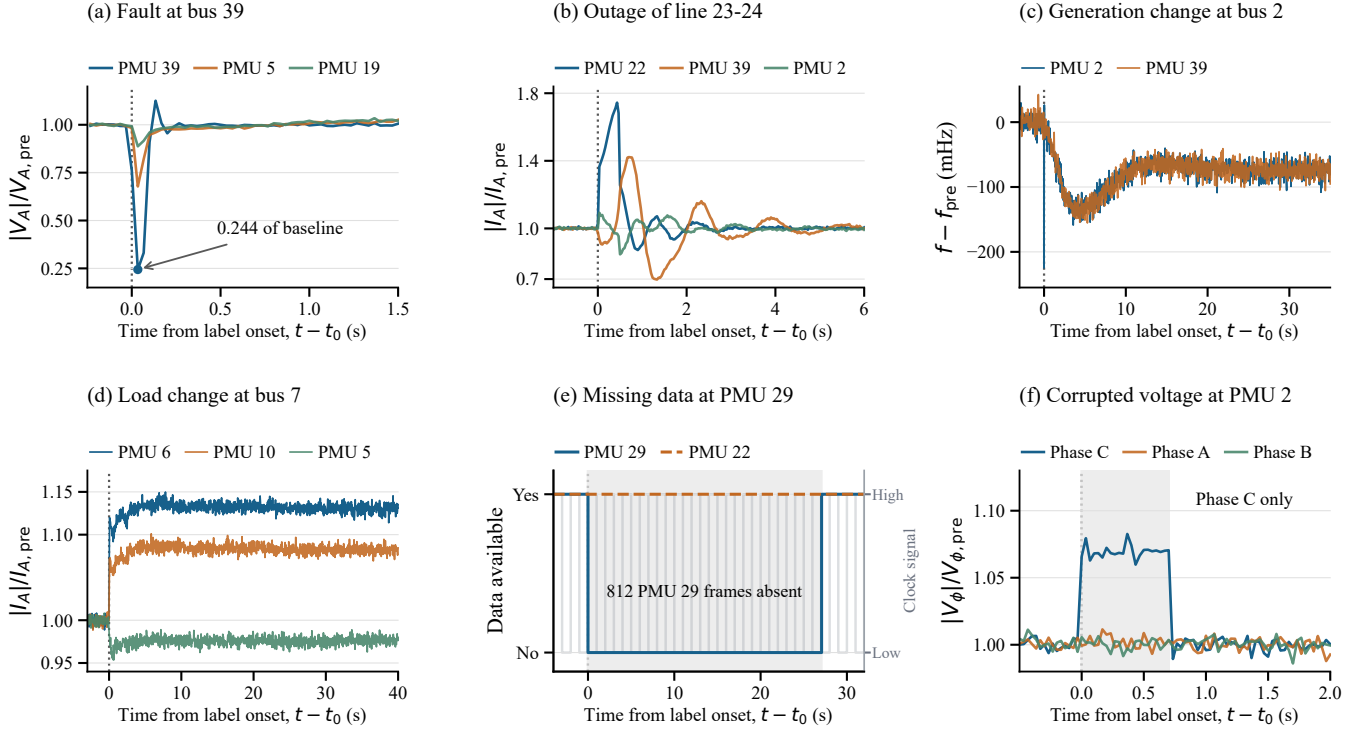
The ordinary split places new replicas of known sources in TEST. RQ4 imposes the harder contract: it removes each physical source from every source-labeled fitting and tuning step and supplies the true event family at evaluation. This isolates physical-source generalization from detection and family-classification errors.

Figure 4 grounds the event ontology before any score is reported. A fault produces a short voltage sag, a line outage redistributes current, and injection changes evolve more slowly. Missing or corrupted measurements require different evidence. The clock trace in panel (e) makes the distinction explicit: time advances while packets from PMU 29 are absent.

All intervals and tests use the experiment’s independent unit: a complete trajectory, event parent, or rebuilt plant. Generated tables and figures read only the hashed frozen

TABLE 3. Four research questions and their evidence contracts. Results from different rows are not pooled.

RQ	Scientific question	Primary evidence and independent unit	Claim boundary
RQ1	What hidden voltage information do eight PMUs recover?	100 frozen nonlinear PowerDynamics TEST trajectories; complete trajectory is the unit	Matched model and fixed PMU set
RQ2	What fails under physical mismatch, and can a constrained recentering repair it?	980 independently rebuilt plants for diagnosis; 60 held-out equilibria for physical recentering	Point recovery under tested shifts, not sequential adaptation
RQ3	When can simultaneous load sources be resolved?	3,488 fresh nonlinear paths and 10,564 complete-record decisions over all 137 hypotheses	One event family, known onset, fixed topology and sensing
RQ4	Does supervised localization survive physical-source holdout?	303 decisions from 101 held-out event parents in ANDES	Historical baseline; true event family supplied

**FIGURE 4.** Representative measured signatures from the RAW0001 reference simulation. Physical events alter voltage, current, or frequency through the network; integrity events alter availability or channel consistency. In panel (e), the gray reference clock advances while PMU 29 packets are absent. Curves are unsmoothed, and each panel uses the time scale needed to show its physical response.

snapshot. Sample-level PMU frames are never promoted to independent replicates.

VII. STATE EVIDENCE FROM RECOVERY TO MODEL FAILURE

A. RQ1 MEASUREMENT CONDITIONING RECOVERS THE TARGET

The estimator sequence reveals where the nominal accuracy comes from. Conditioning the equilibrium on the current PMU frame (B1) reduces hidden-voltage error by more than one order of magnitude. Adding causal propagation (B2) improves most trajectories, but by a much smaller amount. Thus the network-constrained measurement map supplies the main gain; temporal propagation refines it.

Figure 5(a) shows the scale change directly, and panel (c) uses paired trajectory differences rather than overlapping marginal intervals. The comparison is deliberately internal:

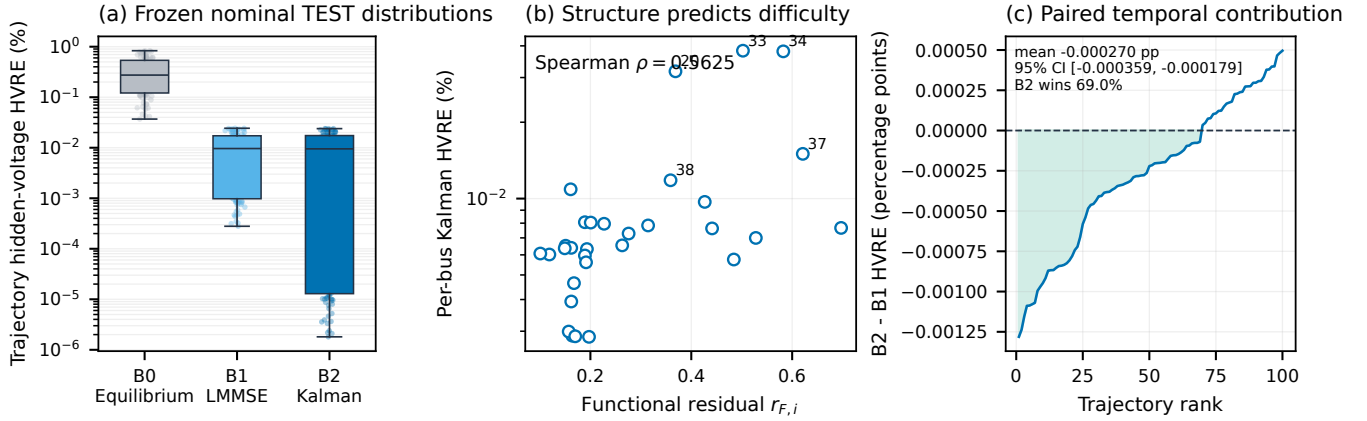
B0, B1, and B2 share the same local physical model. It does not claim superiority over every learned or nonlinear estimator.

The structural residual in Fig. 5(b) correlates with bus-level B2 error, but the relationship is incomplete. The buses share trajectories and network coupling, and the residual comes from the threshold-sensitive projector discussed in Section IV. It is therefore a difficulty indicator, not an observability certificate.

Two negative checks sharpen this result. The fixed-lag smoother does not improve the nonlinear TEST trajectories, even though a correctly specified linear-Gaussian smoother cannot increase expected squared error. Innovation autocorrelation identifies the missing premise: the local model treats deterministic governor and load excitation as white process noise. Marginal temperature scaling repairs coverage under

TABLE 4. Nominal reconstruction on 100 frozen nonlinear TEST trajectories. HVRE is the trajectory-and-bus mean relative complex-voltage error over the 31 unobserved buses; intervals bootstrap independent trajectories.

Method	Complex RMSE	$ V $ RMSE	Angle RMSE ($^\circ$)	Mean HVRE (%)	95% HVRE interval (%)
B0 equilibrium	3.486e-03	2.496e-05	0.19497	0.340200	[0.293231, 0.387428]
B1 snapshot LMMSE	1.529e-04	5.091e-05	0.00821	0.009904	[0.008342, 0.011514]
B2 causal Kalman	1.338e-04	4.235e-05	0.00722	0.009634	[0.008006, 0.011302]

**FIGURE 5.** Where nominal state-reconstruction accuracy comes from. (a) PMU conditioning produces the dominant reduction in trajectory-level hidden-voltage error. (b) The preregistered structural residual predicts some, but not all, difficult buses. (c) Paired B2-minus-B1 differences isolate the smaller contribution of causal propagation.

nominal operation but does not whiten those innovations. More data cannot compensate for the wrong temporal model.

B. RQ2 PHYSICAL SHIFT EXPOSES THE MODEL BOUNDARY

The mismatch campaign asks whether nominal accuracy survives a changed plant. It does not fail uniformly. Network-parameter changes and operating-point shifts dominate the state error, while the tested machine, governor, AVR, and load-model perturbations remain close to the nominal level over the short excitation [Fig. 6(a)]. Those near-null responses show that the experiment did not expose the corresponding parameters; they are not evidence of general robustness.

This pattern suggests a mechanism. A stale equilibrium moves the center of both the measured and hidden voltage fields. The offset-only ablation estimates that movement from measured buses but ignores AC balance, and it closes almost none of the oracle gap. The constrained recentering in (14) instead rebuilds a physically admissible center. Across all three held-out operating shifts, it removes nearly all of the point-error gap [Fig. 7(a)–(b)].

The recovery does not identify every operating-point coordinate. The nuisance information is rank deficient, yet the requested voltage field changes little along much of that nullspace. This is the empirical counterpart of Proposition 1: a target can be recoverable even when the full latent vector is not. The remaining limitation is uncertainty. Marginal intervals cover, but the joint NEES-like statistic is severely under-dispersed, and the independent constrained solver converges on only part of its small diagnostic subset. The evidence

supports point reconstruction for the tested operating shifts, not joint parameter recovery or sequential adaptation.

VIII. EVENT EVIDENCE FROM DETECTION TO SOURCE AMBIGUITY

A. RQ3 CARDINALITY RESOLVES BEFORE EXACT SUPPORT

The prospective test opens the full 137-hypothesis bank only after the physical dictionary, covariance, priors, onset, and numerical integration are frozen. Every source pair competes in one normalized posterior. The resulting pattern is monotone with event strength, but the two decisions do not improve at the same rate (Table 6). Finite pairs are recovered exactly, whereas weak-weak Top-1 remains 24.4%. Weak sources are therefore the decisive regime, not a small perturbation around an otherwise uniform accuracy claim.

Figure 8 follows the inference failure in the same order as the method. Panel (a) establishes the strength-dependent recovery boundary. Panel (b) then shows that evidence for $K = 2$ grows while exact pair recovery remains nearly flat. The model has detected multiplicity before it has localized the support. Reporting only event detection would hide this failure.

Event presence is easier than source recovery. The frozen detector separates most event and no-event records, but its no-event exposure is too short for an operational false-alarm claim. Per-bus inclusion probabilities also rank useful candidates, yet their missed-source rate remains too high to repair weak exact-support decisions. Supplementary Appendix G reports these secondary metrics and denominators.

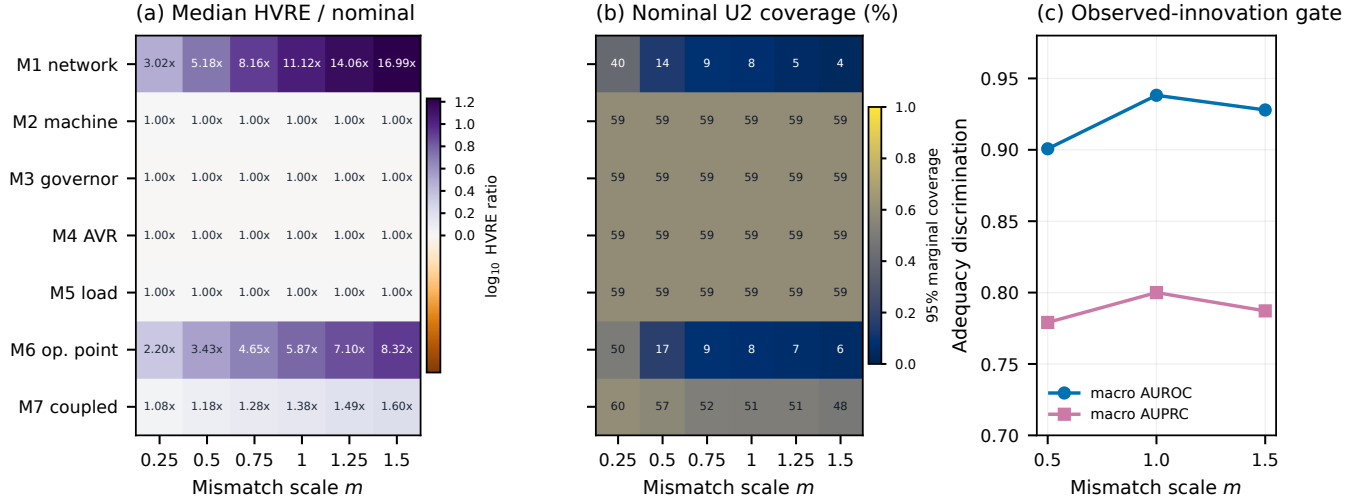


FIGURE 6. How the frozen state estimator fails under physical mismatch. (a) Network and operating-point changes dominate the hidden-voltage error. (b) A covariance correction fitted under nominal operation loses coverage after physical shift. (c) Observed innovations often flag unacceptable hidden error but do not identify its physical cause. Each family-scale cell contains 20 rebuilt plants.

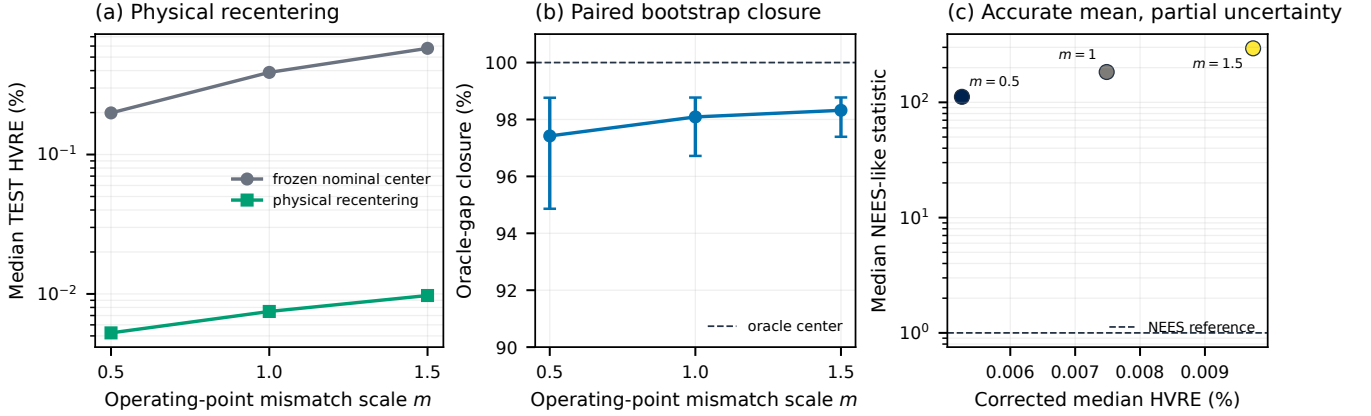


FIGURE 7. Physical recentering separates point recovery from uncertainty. (a) Enforcing the AC map sharply reduces hidden-voltage error under operating-point shift. (b) The paired oracle-gap closure is consistent across held-out cases. (c) The joint NEES-like diagnostic remains too large even when marginal intervals cover, so the point estimate is supported but its joint covariance is not.

TABLE 5. Corrected physical static recentering on 60 held-out M6 operating-point cases. Gap-closure intervals bootstrap the 20 independent TEST cases at each scale. Rank and nullity refer to the 43-dimensional nuisance basis; runtime is per solve.

m	n	Frozen HVRE (%)	Corrected HVRE (%)	Gap closure [95% CI] (%)	Rank	Nullity	Median NEES	Runtime (ms)
0.5	20	0.19885	0.00525	97.4 [94.9, 98.8]	25/43	18	111.6	57.6
1.0	20	0.38834	0.00749	98.1 [96.7, 98.8]	25/43	18	183.3	61.1
1.5	20	0.57795	0.00974	98.3 [97.4, 98.8]	25/43	18	294.2	69.2

B. PROFILED GEOMETRY EXPLAINS THE WEAK CASES

The retrospective audit asks why the weak bank fails. A support-only score compares candidate signatures while holding the rest of the explanation fixed. The profiled distance instead allows the competing support to choose its best amplitude and thereby measures the actual overlap between response manifolds. The latter tracks exact weak-case recovery far more closely [Fig. 8(c)].

This result changes the interpretation of the weak regime. Every tested direction is first-order resolvable, so a curvature-

dominated a^{-4} delay law is not the observed mechanism. The errors arise because several finite-horizon candidate responses lie close after amplitude profiling. Detectability and source identity therefore require different geometric quantities.

The 120-frame diagnostic adds a second piece of the explanation. Extending the window increases the separation proxy by more than eightfold, and no candidate pair is exactly degenerate in the frozen equilibrium model. Yet the hardest signatures remain almost collinear [Fig. 8(e)–(f)]. More time

TABLE 6. Fresh prospective 137-hypothesis confirmation. The bank contains H_0 , 16 single-source hypotheses, and all 120 two-source supports. Repeated noise realizations are decisions, not independent physical trajectories. Top- k is exact support recovery and is undefined for H_0 and single-source rows in this exported support table.

Regime	Physical trajectories	Decisions	Cardinality (%)	Top-1 (%)	Top-3 (%)	Mean $p(K = 2)$
No event	—	100	100.0	—	—	0.000
Single source	128	384	93.8	—	—	0.051
Weak-weak pair	1920	5760	37.4	24.4	59.8	0.427
Weak-strong pair	480	1440	77.6	66.2	95.8	0.808
Moderate pair	480	1440	99.7	98.8	100.0	0.996
Finite pair	480	1440	100.0	100.0	100.0	1.000

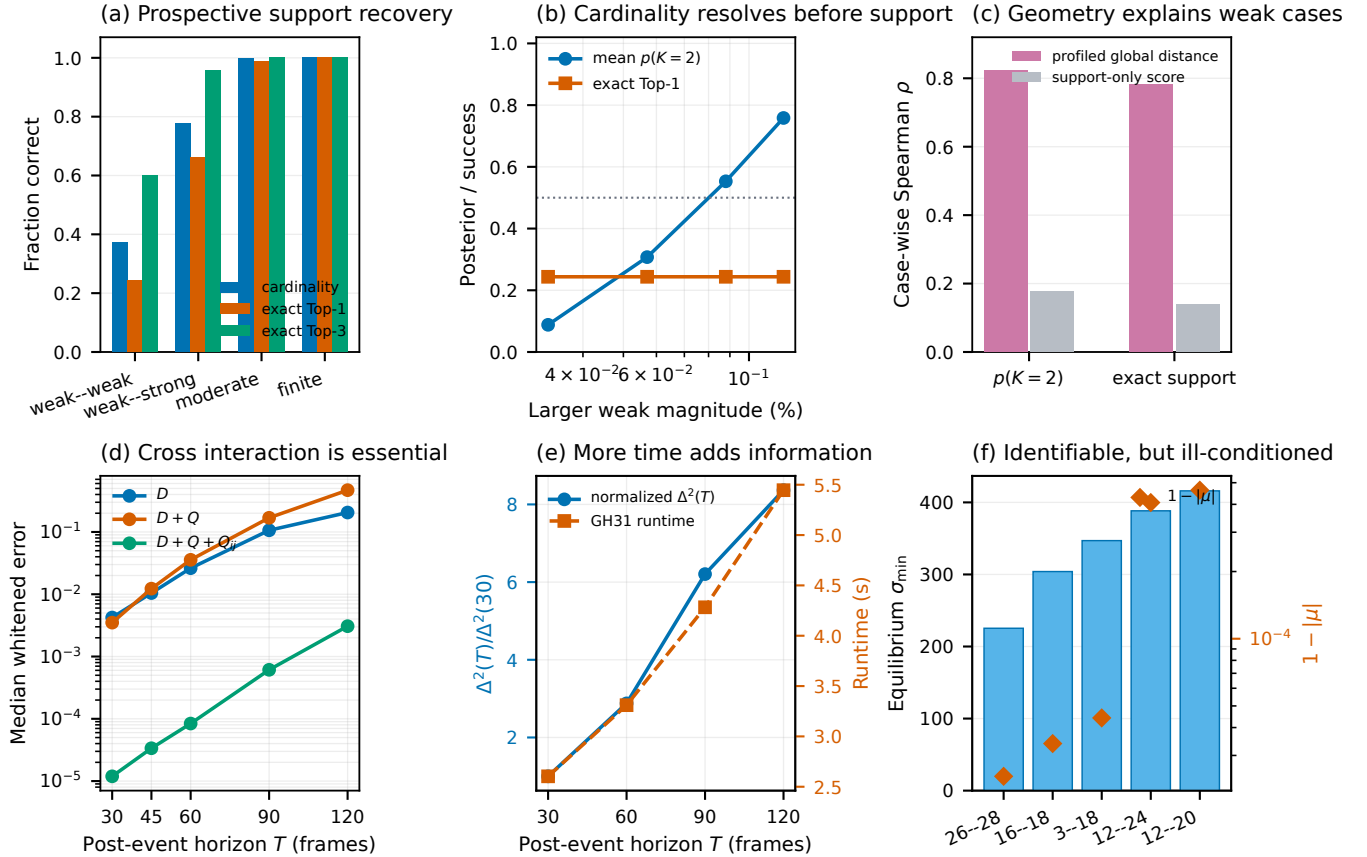


FIGURE 8. Why simultaneous-source recovery fails in the weak regime. Prospective results first separate cardinality from exact support (a–b). Retrospective geometry then links weak-case errors to profiled response distance (c). The longer-horizon diagnostic shows that the pairwise interaction is needed (d), information accumulates with time (e), and distinct equilibrium signatures can remain nearly collinear (f). Panels (c)–(f) explain the result but do not replace a prospective recovery test.

adds information without guaranteeing a well-conditioned decision.

The physical-manifold check also shows why the nonlinear interaction in (16) cannot be dropped. Adding only single-source curvature does not improve the simultaneous response; adding Q_{ij} reduces the median whitened error by roughly two orders of magnitude [Fig. 8(d)]. This check covers 32 nominal-operating-point paths. It validates the representation locally, not population recovery at 120 frames or robustness to operating shift.

C. RQ4 SOURCE LABELS DO NOT GENERALIZE TO NEW ASSETS

The supervised hierarchy provides a useful contrast. On the ordinary split it detects events and ranks known sources well, confirming that multiscale dynamics, adaptive residuals, and PMU contrasts carry event information. That result answers interpolation at represented assets.

The source-holdout result answers a different question. With the physical asset removed from all source-labeled fitting and the true event family supplied, only 3 of 303 decisions are exact. The failure remains after detection and family confusion have been removed. The location heads and asset descriptors therefore memorize source-specific associations

that do not transfer to an unseen physical source.

The physical load bank avoids that shortcut by generating a response for every candidate support. Its success is narrower in another direction: it covers only load interventions under fixed topology, known onset, matched operation, and synthetic noise. Together, the two experiments motivate the next model rather than a claim of general superiority: learned residuals should correct a physical candidate likelihood and must be tested on event parents and source assets excluded from training.

IX. DISCUSSION AND LIMITATIONS

The experiments support one central distinction: recovering a requested state function is not the same as identifying the intervention that produced it. The state estimator can reconstruct hidden voltages because many unobservable state directions have little effect on that target. Source inference is stricter. It must separate every admissible candidate after amplitude and other nuisance directions are allowed to adjust.

The state results make this distinction concrete. Instantaneous PMU conditioning supplies most of the nominal improvement, while causal propagation adds a smaller correction. When the equilibrium becomes stale, an unconstrained offset does not solve the problem; rebuilding an AC-feasible center does. The resulting point estimate is accurate even though the operating-point coordinates remain non-unique. Its joint covariance is still unreliable. Target recovery and parameter recovery must therefore be evaluated separately.

The event results expose a parallel hierarchy. Weak simultaneous events often provide enough evidence for two active sources before they identify the pair. Profiled manifold distance explains this behavior because it measures what remains after a competing support chooses its best amplitude. Longer observation increases that distance, but nearly collinear signatures still amplify noise. A positive structural margin is therefore a uniqueness statement, not a reliability guarantee.

The supervised baseline adds a warning about learning. Engineered temporal and spatial features work when the source appears in training. Under complete source holdout, the same representation loses source identity even when the event family is supplied. A learned correction should therefore operate on candidate residuals, not on asset labels alone, and its value must be measured against the physics-only likelihood on the same event parents.

The evidence remains bounded in four ways. First, the state and event paths are evaluated separately rather than as one fitted posterior. Second, the Bayesian support result covers one event family, fixed topology, known onset, fixed PMUs, and matched operation. Third, the 120-frame study is diagnostic and lacks independent operating-point validation. Fourth, the normal exposure is too short to establish false alarms per hour, and no field or cross-simulator validation is claimed.

The next experiment should close these gaps in that order. It should first freeze the 30–120-frame recovery contract at independent operating points. It should then compare learned-

only, physics-only, residual-corrected, and diagnosability-aware variants on identical source-disjoint parents. Only after those checks should the bank expand to faults, outages, generation changes, integrity nuisance, changing PMU sets, and long continuous streams. This sequence tests the mechanism before adding scope.

X. CONCLUSION

Sparse PMUs impose different information limits on state reconstruction and event-source inference. The requested hidden-voltage function can be estimated accurately even when the full local state is not observable. Event support requires a stronger condition: candidate responses must remain separated after amplitude and nuisance directions are profiled.

The experiments follow that logic. Eight PMUs reconstruct the 31 unmeasured bus voltages with 0.009634% mean relative error under the matched model, and physical recentering repairs most of the tested operating-point shift. In the prospective load bank, finite two-source supports are recovered exactly, whereas weak-pair Top-1 remains 24.4%. The gap is explained by finite-horizon response overlap and near-collinearity, not by event absence alone.

The present evidence supports matched-model state reconstruction and simultaneous load-source inference as separate components. A common multi-family experiment with state and integrity nuisance, source holdout, changing sensing, and physical shift is still required before a joint operational decision can be trusted.

SUPPLEMENTARY MATERIAL

The companion supplement contains the paper-by-paper literature matrix, electrical and PMU conventions, proofs, complete estimator equations, frozen experimental contracts, and secondary result tables. These details preserve reproducibility without interrupting the main scientific argument.

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